Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Lab Activity - 7

Full Marks: 50 *Deadline: 11:59 PM, 27-08-2025*

Submission Instructions:

1) Answer all the questions.
2) Write all the scripts, execute and take screenshots of the scripts, name the scripts files as <Your_Roll_No>_<Lab_Activity_No>_<Question_No>.
3) Prepare a report pdf which contains the question number and screenshot of script code and corresponding outputs. Report quality will contain **5 marks**. You should prepare a zip folder which will contain the report and all the directories and files that you have created during execution of the questions.
4) Naming of the zip file should strictly follow the given format:
   <Roll_No>_<Name>_<Activity_No>
   If your name is Shareen, roll number is B24CS055 then the filename should be:
   B24CS055_Shareen_04.pdf
5) **Plagiarism is strictly prohibited.**

Objectives:

1. To practice using Makefiles and CMake for automating compilation, file handling, and project management.
2. To understand modular programming by separating code into .c and .h files and compiling them together.
3. To apply automation with Bash scripting for running C programs multiple times and processing outputs.

4. To strengthen skills in building automation, scripting, and command-line arguments for real-world software development.

Note for Makefiles : Create separate directory for each question (i.e Q1 to Q5) to avoid conflicts in Makefiles each folder should be named as <Your Roll Number>_<Question Number>

1. Create a Makefile such that on executing the command:
    a. Run ***make create_file*** to create the file (Ex: sample.txt). (Don't create file in prior write the code in the Makefile to create the file)
    b. Run ***make print_file*** to print its contents. (content of the file which is created on executing the above command)
    c. Use ***make clean*** to delete the file. (which is created)

[5 Marks]

```
p4@p4:~/ssl/12th$ make create_file
echo "This is a sample file content" > sample.txt
p4@p4:~/ssl/12th$ make print_file
This is a sample file content
p4@p4:~/ssl/12th$ make clean
rm -f sample.txt
p4@p4:~/ssl/12th$
```

2. Write a Makefile that compiles a program using main.c and pallindrome.c. The main.c file should call a function defined in pallindrome.c, with the function prototype declared in pallindrome.h.

[10 Marks]

```
snsrl1@snsrl:~/SSL_2024/Makefile_recaps/Ex3$ ls
main.c  makefile  pallindrome.c  pallindrome.h
snsrl1@snsrl:~/SSL_2024/Makefile_recaps/Ex3$ make
gcc -Wall -I. -c main.c
gcc -Wall -I. -c pallindrome.c
gcc -Wall -I. -o palindrome_check main.o pallindrome.o
snsrl1@snsrl:~/SSL_2024/Makefile_recaps/Ex3$ ls
main.c  makefile          pallindrome.c  pallindrome.o
main.o  palindrome_check  pallindrome.h
snsrl1@snsrl:~/SSL_2024/Makefile_recaps/Ex3$ ./palindrome_check
Enter a number: 121
121 is a palindrome.
```

3. Write a CMakeList.txt file for the above program and execute it using CMake. [10 Marks]
4. Write a Makefile that compiles a simple C project with the following files:
   - main.c
   - functions.c
   - functions.h

   The final executable should be named as a program. You should not directly subtract or add two numbers. You should pass two numbers to function add and sub whose definition and declaration should be present in the functions.c and functions.h respectively.

   [10 Marks]

```
p4@p4:~/ssl$ make
g++ -Wall -std=c++11 -c main.cpp -o main.o
g++ -Wall -std=c++11 -c functions.cpp -o functions.o
g++ -Wall -std=c++11 -o program main.o functions.o
p4@p4:~/ssl$ ./program
Addition: 8
Subtraction: 2
p4@p4:~/ssl$ make clean
rm -f main.o functions.o program
```

5. Write a C program that will take an argument as input and prints the argument if it is a prime number. Write a bash script to take n as input and automate the c program for each value of 1 to n given as argument to the c file and capture the prime outputs and print through script.

   [10 Marks]